

## Farming technology

Ever since the Agricultural Revolution in the 18th century, scientific advances in agriculture have rapidly raised farms' productivity. Artificial fertilizers, pesticides, research into selective breeding, and genetic engineering keep the increasing Western populations fed. Selective breeding programmes improve crops and livestock by combining the strengths of each parent in their offspring. By modifying genes directly in the laboratory, scientists can achieve the same results as selective breeding in a fraction of the time.

### Chemicals in farming

Fertilizers enrich the soil with the minerals essential for plant growth. Animal manure is an organic fertilizer; science has also developed inorganic (artificial) fertilizers. Pesticides are chemicals that control the pests which attack crops. While these chemicals improve productivity, some chemicals get into the water supply, possibly harming human health.

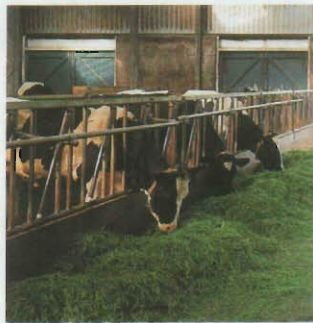
Vegetables can be bred to be regular in shape, and so more attractive to shoppers, even though there is no difference in taste.



Intensively farmed



Irregular shape



Cattle raised in zero-grazing pen

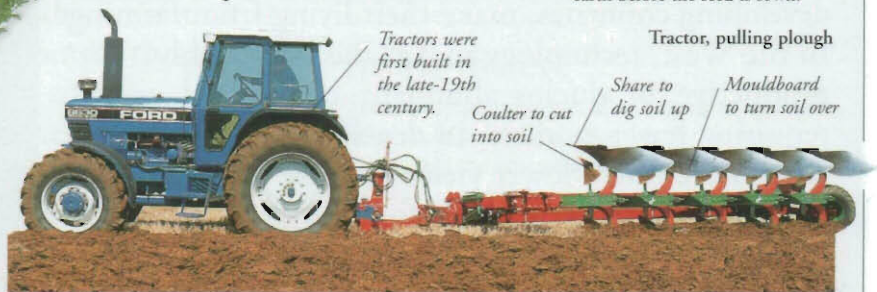
### Intensive rearing

Intensive farming maximizes the production of crops and livestock with chemicals and machines. Shoppers prefer cheap food: to produce this, the cost of raising livestock must be kept to a minimum. Cattle, pigs, and other animals live in controlled conditions; for example, pigs live in units, and are fed a precise mix of nutrients that make them put on the most weight in the least time.

## Machinery

During the 20th century, new machinery increased farm productivity in industrialized nations, while also cutting labour requirements. Tractors, motor vehicles that pull farm machinery, quickly replaced horses as the source of pulling power on the farm.

The tractor's power allowed farmers to plant and harvest crops over vast areas.



Tractor, pulling plough

Because combine harvesters work so quickly, they reap a crop at its best.



### Harvesters

Machines can speed the harvest. A combine harvester both cuts a crop of wheat and separates the grain from the stalk. In less than an hour, a machine with one driver can turn two hectares of wheat into grain. Sugar, potatoes, and peas are other crops harvested by machine.

### Equipment

Milking machines have changed the face of dairy farms, with farmers now able to keep much larger herds of cattle. Farmers operate machines that milk cattle twice a day, although in the Netherlands robotic machines have been introduced that milk cows whenever the cows want it. The robots place vacuum cups on the cows' teats to draw out their milk. Each cow wears a radio collar which tells the robot where to look for the udder.



Robotic milking parlour, the Netherlands

Apples are the main fruit crop in cool, temperate regions. There are more than 6,000 different varieties.

Trees have been pruned so that the Sun can ripen the fruit.

## The farmer's year

The changing seasons set the pace of a farmer's work. On a livestock farm, for instance, spring is the time when lambs or calves are born. The arable (crop-growing) farm is busiest during the spring-to-autumn growing season. Apple growers must carry out different tasks through the year to produce the best crop.

To make mechanical weeding easier, the trees are grown in neat rows.

Winter: the trees take a rest from growing. The farmer prunes the branches into shapes that make the fruit easier to reach.

Bees dust pollen from blossom of one tree to that of another.

Farmer prunes trees.

Beehive

Free-range pigs eat windfall apples.

Spring: bees from hives placed in the orchard in spring pollinate the trees.

Trees are sprayed and fed in spring and summer.

Tractor moves bins of picked apples.

Late summer: the fruits ripen at the end of summer, when harvest (by hand) begins.

Autumn: harvest continues. Windfalls are used to make cider and juice.

Modern apple trees have been bred to be small, making it easier for people to pick the fruit.

